

Self-similar twisted Hilbert extensions are uniformly block-natural centralizer extensions

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Status and source conjecture

This is a candidate partial result, subject to human review. Corrêa, Dantas, and Rodríguez-Vidanes prove that the Kalton–Peck space is the unique nontrivial self-similar twisted Hilbert space among the spaces $Z(\varphi)$ generated by a scalar Lipschitz function, and then conjecture that the same is true for every twisted Hilbert space [1, Theorem 7.4 and p. 19]:

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for every $N, M \geq 1$. Since φ is Lipschitz, we obtain that

$$\sup_{a,b>0} |\varphi(a+b) - \varphi(a) - \varphi(b)| < \infty.$$

Following [31], there exists $c = \lim_{t \rightarrow \infty} \frac{\varphi(t)}{t}$ and

$$\sup_{t>0} |\varphi(t) - ct| < \infty$$

and therefore $Z(\varphi)$ is projectively equivalent to Z_2 . $\square$

As two-thirds of the authors of [11], we conjecture that the Kalton–Peck space Z_2 is, up to isomorphism, the only nontrivial self-similar twisted Hilbert space.

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The exact conjecture is:

The Kalton–Peck space Z_2 is, up to isomorphism, the only nontrivial self-similar twisted Hilbert space.

The theorem below does not finish the classification. It reduces the full problem from arbitrary quasilinear extensions of ℓ_2 by ℓ_2 to extensions generated by exact ℓ_∞ -centralizers. Moreover, the canonical block defects of the resulting centralizer have one bound valid for *all* normalized block bases. The associated scalar indicator therefore satisfies a dimension-free approximate Shannon chain rule for every ordered refinement of a finite probability vector. In particular, a counterexample cannot arise solely from a genuinely non-centralizer quasilinear map, and the remaining classification has a precise scalar cocycle obstruction.

Definitions

We work over $\mathbb{R}$ or $\mathbb{C}$. Let $G = \{-1, 1\}^{\mathbb{N}}$ be the Cantor sign group in either scalar field. For $u \in G$, D_u denotes the diagonal isometry $(x_n) \mapsto (u_n x_n)$ of ℓ_2 .

A homogeneous map $\Omega : c_{00} \rightarrow \ell_2$ is *quasilinear* if there is $Q < \infty$ such that

$$\|\Omega(x + y) - \Omega(x) - \Omega(y)\|_2 \leq Q(\|x\|_2 + \|y\|_2) \quad (x, y \in c_{00}).$$

It represents the exact sequence whose middle space is the completion of $\ell_2 \times c_{00}$ for the quasinorm

$$\|(y, x)\|_\Omega = \|y - \Omega(x)\|_2 + \|x\|_2.$$

Replacing Ω by $\Omega + L + B$, with $L : c_{00} \rightarrow \ell_2$ linear and $B : c_{00} \rightarrow \ell_2$ bounded homogeneous, gives an equivalent exact sequence.

The map Ω is *sign-exact* if

$$\Omega(D_u x) = D_u \Omega(x) \quad (u \in G, x \in c_{00}).$$

In the complex case, *exact* means the same identity for every $u \in \mathbb{T}^{\mathbb{N}}$; over the reals, sign-exact and exact coincide. It is an ℓ_∞ -*centralizer* if there is $C < \infty$ such that

$$\|\Omega(ax) - a\Omega(x)\|_2 \leq C\|a\|_\infty\|x\|_2 \quad (a \in \ell_\infty, x \in c_{00}).$$

This is the standard sequence-space centralizer estimate; see, for example, [2, Sec. 3].

For a normalized block basis (u_n) of the canonical basis of ℓ_2 , write $U = [u_n]$ and let $T_U : \ell_2 \rightarrow U$ be the isometry $T_U e_n = u_n$. The extension $X = X_\Omega$ is self-similar with respect to (u_n) if there is an isomorphism from the original exact sequence to its block restriction inducing T_U on both the kernel and quotient. It is self-similar if this holds for every normalized block basis, as in [1, Sec. 7].

Idea of the proof

Singleton block bases with arbitrary coordinate signs show that the extension class of Ω is fixed by every sign-diagonal conjugacy. This is initially only an algebraic assertion: for every sign sequence there is a possibly different, possibly unbounded linear correction and a bounded error whose bound may depend on the sign sequence. The first Baire argument makes the bounded errors uniform. An invariant mean on the compact sign group can then be applied safely on each finite-support vector, producing an equivalent sign-exact representative ω .

Sign-exactness makes ω support-preserving. To test multiplication by an arbitrary $a \in B_{\ell_\infty}$, encode a_n as the first coordinate of a unit vector in a disjoint two-dimensional block. Self-similarity for this block basis turns the multiplier commutator $a\omega - \omega a$ into a bounded map, up to a bounded diagonal correction. Thus each individual multiplier has a finite commutator bound. The second Baire argument promotes these pointwise finite bounds to one uniform bound over the unit ball of ℓ_∞ , which is exactly the centralizer condition. In the complex case, the centralizer bound then permits a final bounded average over $\mathbb{T}^{\mathbb{N}}$, giving a fully exact representative.

There is a further uniformity mechanism. If the commutator constants for increasing coordinate injections were unbounded, choose successively worse finite witnesses, move their domain and range coordinates into disjoint windows, and paste all of them into one increasing injection. The resulting single commutator would be unbounded, contrary to self-similarity. The same gliding-hump paste, now using the uniform injection bound on both the domain and range, gives one constant for the canonical defects of every normalized block basis. Pairing that vector-valued defect with the corresponding block sum turns it into the approximate chain rule stated below.

The reduction theorem

Theorem 1. *Let*

$$0 \longrightarrow \ell_2 \longrightarrow X \longrightarrow \ell_2 \longrightarrow 0$$

be a self-similar twisted Hilbert extension. Then this exact sequence is equivalent to one generated by a quasilinear map $\omega : c_{00} \rightarrow \ell_2$ satisfying

- (i) $\omega(e_n) = 0$ for every n ;
- (ii) $\omega(D_u x) = D_u \omega(x)$ for every unimodular diagonal u (all $u \in \mathbb{T}^{\mathbb{N}}$ over $\mathbb{C}$, and all signs over $\mathbb{R}$) and $x \in c_{00}$;
- (iii) *there is $C < \infty$ such that*

$$\|\omega(ax) - a\omega(x)\|_2 \leq C\|a\|_{\infty}\|x\|_2 \quad (a \in \ell_{\infty}, x \in c_{00}).$$

Thus every self-similar twisted Hilbert extension is centralizer-generated.

The proof also gives the following strengthening. It is stated separately so that the scalar obstruction left by the source conjecture is explicit.

Theorem 2 (Uniform block naturality and indicator chain rule). *The representative ω in Theorem 1 can be chosen so that there is a constant $K < \infty$, independent of the normalized block basis $U = (u_n)$, satisfying*

$$\left\| T_U \omega(x) - \omega(T_U x) + \sum_n x_n \omega(u_n) \right\|_2 \leq K\|x\|_2 \quad (x \in c_{00}). \quad (1)$$

For every finite positive vector $p = (p_n)$ put

$$\mathcal{H}_{\omega}(p) = \langle \omega((\sqrt{p_n})_n), (\sqrt{p_n})_n \rangle.$$

If $q^{(n)} = (q_{n,k})_k$ are finite probability vectors and $r = (p_n q_{n,k})_{n,k}$ is their ordered concatenated refinement, then

$$\left| \mathcal{H}_{\omega}(r) - \mathcal{H}_{\omega}(p) - \sum_n p_n \mathcal{H}_{\omega}(q^{(n)}) \right| \leq K \sum_n p_n. \quad (2)$$

Here the finite vectors may be placed in arbitrary successive coordinate blocks; changing their placement changes only the same uniform error class.

We divide the proof into lemmas.

Lemma 3 (Block commutator criterion). *Let $\Omega : c_{00} \rightarrow \ell_2$ represent the extension and let (u_n) be a normalized block basis. If the extension is self-similar with respect to (u_n) , then there are a linear map $L : c_{00} \rightarrow U$ and $C < \infty$ such that*

$$\|T_U \Omega(x) - \Omega(T_U x) + Lx\|_2 \leq C\|x\|_2 \quad (x \in c_{00}).$$

Proof. Any operator making the defining exact-sequence diagram commute has, on the dense algebraic model, the triangular form

$$T(y, x) = (T_U y + Lx, T_U x)$$

for a linear $L : c_{00} \rightarrow U$. Apply boundedness of T to $(\Omega(x), x)$, whose Ω -quasinorm is $\|x\|_2$. The first component of the resulting target quasinorm is precisely the expression in the statement. This is the general part of the calculation used in [1, Lemma 7.1]. $\square$

First subtract the linear map determined by $e_n \mapsto \Omega(e_n)$. We may and do assume from now on that

$$\Omega(e_n) = 0 \quad (n \in \mathbb{N}). \quad (3)$$

Lemma 4 (Uniform diagonal equivalence). *There is $M < \infty$ such that, for every $u \in G$, one can find a linear map $L_u : c_{00} \rightarrow \ell_2$ with*

$$\|D_u \Omega(x) - \Omega(D_u x) + L_u x\|_2 \leq M \|x\|_2 \quad (x \in c_{00}). \quad (4)$$

Proof. For $m \in \mathbb{N}$, let E_m be the set of $u \in G$ for which such an L_u exists with bound m . The singleton block basis $(u_n e_n)$ and Lemma 3 give $G = \bigcup_m E_m$.

We claim that E_m is closed in the product topology of G . Suppose $u^{(j)} \rightarrow u$ and choose L_j with error at most m . By (3), the diagonal commutator vanishes at every e_n ; hence $\|L_j e_n\|_2 \leq m$. A diagonal weak-subsequence argument in the reflexive space ℓ_2 gives vectors $v_n \in \ell_2$ such that $L_j e_n \rightarrow v_n$ weakly for every n . Define the linear map L on c_{00} by $L e_n = v_n$. For each fixed $x \in c_{00}$, product convergence in the sign group implies $D_{u^{(j)}} x = D_u x$ eventually. The remaining diagonal action on the fixed vector $\Omega(D_u x) \in \ell_2$ converges strongly. It follows that the diagonal commutator converges in norm. Passing to the weak limit in the error inequality yields the same inequality with L and bound m . Thus $u \in E_m$.

The sets satisfy

$$E_m^{-1} \subset E_m, \quad E_m E_k \subset E_{m+k},$$

because diagonal conjugacies are isometries and the corresponding linear corrections may be conjugated and added. Baire category gives an m for which E_m has nonempty interior. Therefore $E_m^{-1} E_m$ contains an open neighborhood W of the identity and $W \subset E_{2m}$. The subgroup generated by W is open in the compact group G , hence has finite index; compactness also gives an integer r for which that subgroup is contained in W^r . Choose finitely many coset representatives and finite bounds for them. The product rule then gives one bound M valid on all of G . $\square$

Lemma 5 (Sign exactification). *The map Ω is equivalent to a sign-exact quasilinear map $\omega : c_{00} \rightarrow \ell_2$ satisfying $\omega(e_n) = 0$.*

Proof. Rewrite (4), after multiplying by D_u^{-1} , as

$$D_u^{-1} \Omega(D_u x) - \Omega(x) = A_u x + B_u(x),$$

where A_u is linear and $\|B_u(x)\|_2 \leq M \|x\|_2$. The sign convention is immaterial. Since the left side vanishes at e_n , we may choose the decomposition so that $\|A_u e_n\|_2 \leq M$. Consequently, for every fixed $x \in c_{00}$, both $u \mapsto A_u x$ and $u \mapsto B_u(x)$ are bounded ℓ_2 -valued functions.

Let $\mathfrak{m}$ be an invariant mean on the compact abelian group G and define weakly in the reflexive space ℓ_2

$$\omega(x) = \mathfrak{m}_u [D_u^{-1} \Omega(D_u x)].$$

The uniform quasilinearity constant of the conjugates shows that ω is quasilinear. Invariance of the mean gives

$$\omega(D_v x) = D_v \omega(x) \quad (v \in G).$$

Moreover,

$$\omega - \Omega = \mathfrak{m}_u A_u + \mathfrak{m}_u B_u,$$

the first summand being linear on c_{00} and the second bounded by M . Thus ω is equivalent to Ω . Finally $\omega(e_n) = 0$ follows from (3). $\square$

Sign-exactness implies the support property

$$\text{supp } \omega(x) \subset \text{supp } x. \quad (5)$$

Indeed, choose a diagonal unit equal to 1 on $\text{supp } x$ and different from 1 off that support, and compare $\omega(ux)$ with $u\omega(x)$.

Lemma 6 (Canonical block correction). *Let (u_n) be a normalized block basis. There is $C_U < \infty$ such that*

$$\left\| T_U \omega(x) - \omega(T_U x) + \sum_n x_n \omega(u_n) \right\|_2 \leq C_U \|x\|_2 \quad (x \in c_{00}). \quad (6)$$

Proof. By Lemma 3, the first two terms become bounded after adding a linear $L : c_{00} \rightarrow U$. Average L over the diagonal group on the domain, acting on the range by multiplication of the whole n th block by the same scalar. Sign-exactness of ω shows that the averaged linear map Λ still gives a bounded correction and that Λe_n is supported in the block supporting u_n .

At e_n , the uncorrected commutator equals $-\omega(u_n)$. Hence $\|\Lambda e_n - \omega(u_n)\|_2$ is bounded independently of n . Because these differences have disjoint supports, the linear map sending e_n to $\Lambda e_n - \omega(u_n)$ is bounded on ℓ_2 . Replacing Λe_n by $\omega(u_n)$ therefore changes only the final constant and yields (6). $\square$

Lemma 7 (Uniform increasing-injection commutators). *For every increasing injection $\alpha : \mathbb{N} \rightarrow \mathbb{N}$, let $J_\alpha e_n = e_{\alpha(n)}$. There is a constant $D < \infty$, independent of α , such that*

$$\|J_\alpha \omega(x) - \omega(J_\alpha x)\|_2 \leq D \|x\|_2 \quad (x \in c_{00}). \quad (7)$$

Proof. For a fixed α , Lemma 6 applied to the singleton block basis $(e_{\alpha(n)})$ gives a finite bound because $\omega(e_{\alpha(n)}) = 0$. Denote the least such bound by $d(\alpha)$.

Suppose that $\sup_\alpha d(\alpha) = \infty$. We construct one increasing injection β on successive finite windows. At a given stage choose a fresh domain shift s and a fresh range shift t so large that a finite increasing prescription beyond the preceding windows can be extended to an increasing injection. Let $\tau_t(n) = t + n$. Since $d(\tau_t) < \infty$, choose an increasing injection α and a unit vector $x \in c_{00}$ such that

$$\|J_\alpha \omega(x) - \omega(J_\alpha x)\|_2 > d(\tau_s) + d(\tau_t) + j$$

at the j th stage. On $\text{supp } x$ prescribe

$$\beta(s + n) = t + \alpha(n).$$

The prescribed source and target values have the same order. By taking the two new windows after the preceding ones and filling the gaps monotonically, all the finite prescriptions extend to one increasing injection β .

Exactness makes ω support-preserving, so the prescription applies both to x and to $\omega(x)$. Writing $J_s e_n = e_{s+n}$, on the shifted witness we therefore have

$$\begin{aligned} J_\beta \omega(J_s x) - \omega(J_\beta J_s x) &= J_\beta (\omega(J_s x) - J_s \omega(x)) \\ &\quad + J_t (J_\alpha \omega(x) - \omega(J_\alpha x)) \\ &\quad + (J_t \omega(J_\alpha x) - \omega(J_t J_\alpha x)). \end{aligned}$$

The first and third terms have norms at most $d(\tau_s)$ and $d(\tau_t)$, while the middle one has the norm of the chosen bad commutator. Hence the commutator for the single injection β has norm greater than j on a unit vector for every j . This contradicts $d(\beta) < \infty$ and proves the lemma. $\square$

Lemma 8 (Uniform canonical block bound). *There is one constant K for which (1) holds for every normalized block basis.*

Proof. Let $\kappa(U)$ be the least constant in (6). Suppose that $\sup_U \kappa(U) = \infty$, and let D be given by Lemma 7. Choose normalized block bases U^j and unit finite-support witnesses x^j for which the left side of (6) is larger than $j + 3D$.

Move the finitely many blocks of U^j used by x^j into a fresh range window, and move the corresponding coefficient coordinates into a fresh domain window. Include the intervening blocks of U^j and fill all other gaps with normalized singleton blocks. Doing this successively produces a single normalized block basis $V = (v_n)$. If J_s and J_t are the increasing domain and range embeddings used for one such window, then on the relevant finite coordinates

$$T_V J_s = J_t T_{U^j}.$$

Put

$$K_U(x) = T_U \omega(x) - \omega(T_U x) + \sum_n x_n \omega(u_n).$$

Using the displayed intertwining identity and adding and subtracting $J_t K_{U^j}(x^j)$ gives

$$\begin{aligned} K_V(J_s x^j) - J_t K_{U^j}(x^j) &= -T_V(J_s \omega(x^j) - \omega(J_s x^j)) \\ &\quad + (J_t \omega(T_{U^j} x^j) - \omega(J_t T_{U^j} x^j)) \\ &\quad - \sum_n x_n^j (J_t \omega(u_n^j) - \omega(J_t u_n^j)). \end{aligned}$$

The first two terms have norm at most D . The vectors in the last sum have disjoint supports, each has norm at most D , and hence that sum also has norm at most $D \|x^j\|_2 = D$. Thus

$$\|K_V(J_s x^j)\|_2 > j.$$

This makes the canonical defect for the single block basis V unbounded, contradicting Lemma 6. $\square$

Proof of Theorem 2. The vector estimate is Lemma 8. For the scalar estimate, by homogeneity it is enough to consider $\sum_n p_n = 1$. Put $x_n = \sqrt{p_n}$ and place $u_n = (\sqrt{q_{n,k}})_k$ in successive disjoint blocks. Then $T_U x = (\sqrt{p_n q_{n,k}})_{n,k}$. Pair (1) with this unit vector. Disjointness gives

$$\begin{aligned} \langle T_U \omega(x), T_U x \rangle &= \mathcal{H}_\omega(p), \\ \langle \omega(T_U x), T_U x \rangle &= \mathcal{H}_\omega(r), \\ \left\langle \sum_n x_n \omega(u_n), T_U x \right\rangle &= \sum_n p_n \mathcal{H}_\omega(q^{(n)}). \end{aligned}$$

Cauchy–Schwarz yields (2); homogeneity restores the factor $\sum_n p_n$. In the complex case the final exactification in Lemma 11 is a bounded support-preserving perturbation. Its three terms in (1) are uniformly bounded (the last one by disjointness), so the same conclusions hold for the fully exact representative after enlarging K . $\square$

Lemma 9 (Each multiplier has a bounded commutator). *For every $a \in \ell_\infty$, the homogeneous map*

$$C_a(x) = a\omega(x) - \omega(ax)$$

is bounded from $(c_{00}, \|\cdot\|_2)$ to ℓ_2 .

Proof. By homogeneity it suffices to take $\|a\|_\infty \leq 1$. Put $b_n = (1 - |a_n|^2)^{1/2}$ and form the normalized block basis

$$u_n = a_n e_{2n-1} + b_n e_{2n}.$$

Let $Jx = \sum_n x_n e_{2n-1}$ and $Kx = \sum_n x_n e_{2n}$, so that $T_U x = J(ax) + K(bx)$. Apply (6) and then project onto the odd coordinates. The singleton block bases (e_{2n-1}) and (e_{2n}) are also covered by (6); their canonical corrections vanish because $\omega(e_n) = 0$. Consequently

$$J^{-1}P_{\text{odd}}\omega(Jz) = \omega(z) + O(\|z\|_2), \quad P_{\text{odd}}\omega(Kz) = 0,$$

where the second identity uses (5).

Quasilinearity applied to $J(ax) + K(bx)$ now turns the odd-coordinate projection of (6) into

$$\|a\omega(x) - \omega(ax) + d_a x\|_2 \leq C'_a \|x\|_2, \quad (8)$$

where d_a is the diagonal sequence obtained from the odd coordinates of $\omega(u_n)$.

This diagonal correction is bounded. Indeed, by quasilinearity, sign-exactness, and $\omega(e_j) = 0$,

$$\|\omega(u_n)\|_2 \leq Q(|a_n| + b_n) \leq \sqrt{2}Q.$$

Thus $\|d_a\|_\infty \leq \sqrt{2}Q$, and (8) proves that C_a is bounded. $\square$

Lemma 10 (Uniform multiplier bound). *There is $C < \infty$ such that*

$$\|C_a(x)\|_2 \leq C\|a\|_\infty \|x\|_2 \quad (a \in \ell_\infty, x \in c_{00}).$$

Proof. Set

$$q(a) = \sup_{0 \neq x \in c_{00}} \frac{\|C_a(x)\|_2}{\|x\|_2}.$$

Lemma 9 says that $q(a) < \infty$ for every $a \in \ell_\infty$. Homogeneity and quasilinearity give

$$q(a + c) \leq q(a) + q(c) + Q(\|a\|_\infty + \|c\|_\infty). \quad (9)$$

Let B be the closed unit ball of ℓ_∞ and $F_m = \{a \in B : q(a) \leq m\}$. These sets need not be closed, but their closures have a controlled loss:

$$\overline{F_m} \subset \{a \in B : q(a) \leq m + Q\}. \quad (10)$$

To prove this, let $a_j \rightarrow a$ in ℓ_∞ with $q(a_j) \leq m$ and fix $x \in c_{00}$. On each fixed finite-dimensional coordinate subspace, ω is bounded: repeated use of quasilinearity and $\omega(e_n) = 0$ gives $\|\omega(z)\|_2 \leq K_{\text{supp } x} \|z\|_2$. Hence $\omega((a - a_j)x) \rightarrow 0$. Quasilinearity applied to $ax = a_j x + (a - a_j)x$ then yields

$$\|C_a(x)\|_2 \leq (m + Q)\|x\|_2$$

after taking the limit. This proves (10).

Since $B = \bigcup_m F_m \subset \bigcup_m \overline{F_m}$, Baire category and (10) give a norm ball $B(a_0, r)$ contained in B on which q is bounded by some M . We may choose a_0 in the norm interior of B . If $\|h\|_\infty < r$, then both a_0 and $a_0 + h$ lie in a slightly smaller such ball. Applying (9) to $h = (a_0 + h) - a_0$ gives a uniform bound for $q(h)$ on a neighborhood of zero. Homogeneity now gives $q(a) \leq C\|a\|_\infty$ for all a . $\square$

Lemma 11 (Complex exactification). *In the complex case, a sign-exact ℓ_∞ -centralizer ω is strongly equivalent to an exact ℓ_∞ -centralizer $\tilde{\omega}$ satisfying $\tilde{\omega}(e_n) = 0$.*

Proof. Let $K = \mathbb{T}^{\mathbb{N}}$. The centralizer estimate gives, uniformly for $u \in K$,

$$\|D_u^{-1}\omega(D_u x) - \omega(x)\|_2 \leq C\|x\|_2.$$

Thus the weak invariant mean

$$\tilde{\omega}(x) = \mathfrak{m}_{u \in K}[D_u^{-1}\omega(D_u x)]$$

is well-defined in the reflexive space ℓ_2 , differs from ω by a bounded homogeneous map, and is exact by invariance of the mean. It remains an ℓ_∞ -centralizer because bounded perturbations preserve the centralizer estimate. The equality on the basis is immediate. $\square$

Proof of Theorem 1. Normalize the original representative on the basis, apply Lemmas 4 and 5, and retain self-similarity because equivalent quasilinear maps represent equivalent exact sequences. Lemmas 6–10 then show that the sign-exact representative ω satisfies the uniform ℓ_∞ -centralizer estimate. Over $\mathbb{R}$ this is already the assertion. Over $\mathbb{C}$, apply Lemma 11. $\square$

What remains open

The two theorems remove one layer of generality from the source conjecture: every self-similar twisted Hilbert extension is represented by an exact centralizer with a uniform block-natural defect, and its scalar indicator satisfies the dimension-free ordered chain rule (2). They do *not* prove that this centralizer is symmetric under coordinate permutations, nor that it has the scalar form

$$x_n \varphi\left(\log \frac{\|x\|_2}{|x_n|}\right)$$

treated in [1]. The remaining classification problem can therefore be stated sharply as follows:

Classify the exact ℓ_∞ -centralizers on ℓ_2 whose induced extension is self-similar with respect to every normalized block basis. Must every nontrivial one generate a space isomorphic to Z_2 ?

The tempting next step “self-similar centralizer implies symmetric centralizer” is not established here. Equivalently at the indicator level, one would like a stable characterization of Shannon entropy from the ordered refinement rule (2) without assuming permutation symmetry. Standard entropy-stability results impose approximate symmetry, whereas block bases preserve the order of their disjoint supports. No valid uniform permutation or ordered-cocycle classification argument was found.

Verification notes

The proof was checked against the following failure modes.

- The first Baire argument does not assume a priori uniform norms for the block witnesses; the closed sets E_m and their product law supply that uniformity.
- The linear corrections may be unbounded on ℓ_2 . They are only averaged on c_{00} , where the uniform column bound makes every orbit bounded; the bounded nonlinear errors are averaged separately.

- Sign-exactness, rather than an unjustified coordinate formula, supplies the support property and the blockwise diagonalization in Lemma 6.
- The uniform block constant is not obtained by transporting blocks with an arbitrary permutation. Lemmas 7 and 8 use only increasing domain and range embeddings and paste finite witnesses into one legitimate block basis.
- The second Baire argument does not claim that the sublevel sets of q are closed. The controlled closure loss (10) is the point that makes the argument valid without continuity of ω on the whole unit sphere.

No computational verification is used. Bounded searches on July 21–22, 2026 of the run’s 2000–2026 local corpus and web/arXiv results for the exact conjecture, “self-similar twisted Hilbert”, “block centralizer”, and arXiv:2408.07827 found the source paper and background papers on centralizers and group actions, but no later resolution or explicit statement of the reduction or the uniform ordered chain rule above. Novelty confidence is therefore moderate, while the mathematical verdict is *likely valid, expert review requested*.

References

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